## What is Cloud Controller Manager?

**Professional Answer**

Cloud Controller Manager integrates Kubernetes with cloud-provider APIs for managing load balancers, volumes, and node lifecycle.

**Unprofessional Answer**

It manages cloud things.

## Why is cloud-controller-manager used in Kubernetes?

cloud-controller-manager integrates Kubernetes with cloud provider services.
It manages cloud-specific resources like load balancers, volumes, and node lifecycle.
This separation keeps Kubernetes core cloud-agnostic.
It communicates directly with cloud provider APIs such as AWS, GCP, or Azure.
It allows Kubernetes to run consistently across different cloud environments.

## cloud-controller-manager (Cloud Internals)

**Why it exists**

- Separates cloud logic from core Kubernetes

**Internal Controllers**

1. **Node Controller**
    - Cloud node lifecycle
2. **Service Controller**
    - Creates cloud load balancers
3. **Route Controller**
    - Manages network routes
4. **Volume Controller**
    - Attaches/detaches disks

**Works with**

- AWS APIs
- GCP APIs
- Azure APIs

## What is Cloud Controller Manager?

Cloud Controller Manager is a **Kubernetes control-plane component** that integrates Kubernetes with **cloud provider APIs**.
It allows Kubernetes to manage cloud resources **without hard-coding cloud logic into core components**.
CCM runs cloud-specific controllers.
It communicates with both **kube-apiserver** and **cloud APIs**.
Without CCM, Kubernetes cannot manage cloud-native resources properly.

## Why Cloud Controller Manager was introduced?

Earlier, cloud logic was inside:

- kube-controller-manager
- kubelet

This caused:

- Tight cloud coupling
- Hard upgrades
- Security risks

CCM **decouples cloud logic** from Kubernetes core.

This is a **design evolution question** interviewers love.

## Where does CCM run?

- As a **control plane component**
- Usually as a **pod** (static pod or deployment)
- Runs only in **cloud environments**
- Not needed for bare-metal clusters

Managed services (EKS, GKE, AKS) run it for you.